

UNIT III MICRONUTRIENTS-VITAMINS AND MINERALS

Fat-soluble Vitamins - Function, RDA, sources and deficiency and excess.

Water soluble vitamins: Thiamin, Riboflavin, Niacin, B12, Folic acid, Biotin and Vitamin C: functions, RDA, food sources, deficiencies.

Macro minerals and Micro Minerals - functions, absorption, RDA, sources and deficiency.

Protective and regulatory foods

- Vitamins and Minerals together form the category of protective and regulatory foods.
- As the name suggests it means that these two categories of nutrients in food protect the body from diseases and infections and also help it to function optimally.

- Vitamins are classified as water-soluble and fat-soluble vitamins depending on their solubility.
- Fat-soluble vitamins-vitamin A, D, E and K dissolve in fat before they are absorbed in the blood stream to carry out their functions. Excesses of these vitamins are stored in the liver. Because they are stored, they are not needed every day in the diet.
- By contrast, water-soluble vitamins dissolve in water and are not stored. They are eliminated in urine. We need a continuous supply of them in our diets. The water-soluble vitamins are the B-complex group and vitamin C

Vitamin A

- The chemical name of vitamin A is retinol. Retinol is referred to as pre-formed vitamin A, as it is present in foods already in the active form and does not require any conversion.
- Retinol itself is found only in animal foods but milk and some vegetable foods also contain the deep yellow or orange carotenes which can be converted in the body to retinol and are therefore sources of vitamin A activity.
- The most important of these is beta carotene which is the primary and most efficient pro vitamin A or carotenoid.

Sources

- Vitamin A or retinol is found only in foods of animal origin, such as milk, cheese, cream, butter, ghee, egg, fish, kidney and liver, liver oils of fish such as halibut, cod and shark.
- Provitamin A (so called because it is a precursor and has to be chemically transformed into retinal) or B-carotene is found primarily in plant foods, which contain orange or yellow colored pigments called carotenoids. B-carotene is the most widely distributed carotenoid in plant foods.
- Palm fruit and red palm oil are the richest source of beta-carotene and dark green leafy vegetables, ripe fruits such as mango, papaya, apricots and yellow orange vegetables like carrot, pumpkin and sweet potato are rich in B-carotene.

Functions

- Vitamin A (retinol) is an essential nutrient needed in small amounts by humans for the normal functioning of the visual system, growth and development, and maintenance of epithelial cellular integrity, immune function, and reproduction.

- Role in growth and cellular differentiation:
- The growth and differentiation of epithelial cells throughout the body are especially affected by vitamin A deficiency. In addition, goblet cell numbers are reduced in epithelial tissues and as a consequence, mucous secretions diminish.
- Cells lining protective tissue surfaces fail to regenerate and differentiate; hence they flatten and accumulate keratin. Xerosis and xerophthalmia are such diseases due to deficiency of vitamin A.

- Integrity of epithelial tissues
- Certain epithelial cells secrete mucous to keep the skin, eyes and other mucous membranes moist.
- Vitamin A is essential for the integrity of the mucous-secreting cells
- Vitamin A maintains the health of epithelial cells that line internal and external surfaces of the lungs, intestines, stomach, vagina, urinary tract and bladder, eyes and skin. These cells act as important barriers to bacteria.
- In deficiency, the epithelial tissues are keratinized. The tissues affected are salivary glands, respiratory tract, eyes, skin.

- Role as antioxidant:
- Vitamin A and related nutrients may collectively be important in protecting against conditions related to oxidative stress, such as aging, air pollution, arthritis, cancer, cardiovascular disease, cataracts, diabetes mellitus and infection.
- Antioxidants protect our cells against the effects of free radicals, are potentially damaging compounds produced as by-products of metabolism, as well as, through exposure to toxins and pollutants (e.g. smoking).

- Role in protein metabolism and growth: Severe vitamin A deficiency results in abnormal RNA metabolism and protein synthesis and hence interferes with growth. Hence vitamin A is also called growth vitamin.
- Role in reproduction: Deficiency of Vitamin A leads to infertility in the male and failure of the female to conceive or resorption or abortion of the fetus.

- Nutritional requirement and RDA for Vitamin A
- In calculating the safe intake, a normative storage requirement equivalent to 434 mg was taken into consideration as anything more than that could also lead to toxicity.

- Health aspects
- Deficiency:
- Vitamin A deficiency (VAD), leads to impairment in the vision, severe infections and even death.
- The disorders are known to include reduced immune competence resulting in increased morbidity and mortality (largely from increased severity of infectious diseases), night blindness, corneal ulcers, keratomalacia and related ocular signs

- Toxicity:
- Because vitamin A is fat-soluble and can be stored, primarily in the liver, routine consumption of large amounts of vitamin A over a period of time can result in toxic symptoms, including liver damage, bone abnormalities and joint pain, headaches, vomiting and skin desquamation.
- In fact, symptoms that occur due to intakes in excess of those recommended over a prolonged period are referred to as symptoms of hypervitaminosis.

- Vitamin D
- Vitamin D is a generic term and can either be made in the skin from a cholesterol-like precursor (7-dehydrocholesterol) by exposure to sunlight or can be provided pre-formed in the diet.
- The version made in the skin is referred to as vitamin D whereas the dietary form can be either vitamin D3 or a closely-related molecule of plant origin known as vitamin D2.

- Food Sources of Vitamin D
- Vitamin D, also called the sunshine vitamin is easily manufactured in the skin from 7-dehydro cholesterol on exposure to sunlight, as already mentioned above .
- Small amounts are present in dairy products such as milk, cheese, butter , and cream, egg yolk, liver , oysters and certain varieties of fish and mushrooms.
- So we have seen that it is not just through diet, sunlight can also help us to manufacture vitamin D.

- Fortified Plant-Based Foods
- Many plant-based products are fortified with vitamin D to help consumers meet their daily requirements.
- Plant-based milks: Soy milk, almond milk, oat milk, and rice milk are often fortified with levels comparable to cow's milk (around 100-140 IU per cup).
- Fortified orange juice, Fortified breakfast cereals and grains: Many cereals and brands of oatmeal are fortified with vitamin D.
- Absorption: Vitamin D is fat-soluble, so consuming these foods or supplements with healthy fats (like nuts, seeds, or avocado) can improve absorption.

- Functions
- Vitamin D is required to maintain normal blood levels of calcium and phosphate, which are in turn needed for the normal mineralization of bone, muscle contraction, nerve conduction and general cellular functions in all cells of the body.
- Mobilization of bone calcium and phosphorus: The functions of vitamin D are mediated by this vital vitamin D hormone by a homeostatic mechanism which involves the hormone acting on the intestines, kidney and bone to increase serum calcium and phosphorus levels. This helps to achieve a normal blood calcium concentration

- Mineralization and formation of new bone: plays a role in the synthesis of a prominent non collagenous protein, osteocalcin, a vitamin K dependent protein found in the bone matrix and dentine - which is associated with new bone formation.
- Bone growth and development-calcification of osteoid tissue: Vitamin D participates in metabolic processes associated with bone growth and development. It is involved in calcification of osteoid tissues. Osteoid is a protein mixture which is secreted by osteoblasts. When it mineralizes, it becomes bone.

- Formation of enzymes: Vitamin D is essential for the formation of two enzymes in the intestinal lining (involved in calcium transport) and adenosine triphosphatase, (for collagen formation in bone matrix).
- Inhibition of cancer cell proliferation and growth: Vitamin D diminishes proliferation of abnormal intestinal, lymphatic, mammary and skeletal cells and provides a potential for the treatment of skin diseases such as psoriasis
- Role in the immune system: Immune responses that are mediated by T-cells can be inhibited by large doses of calcitriol. It is a natural steroid hormone formed in the healthy body as the biologically active form of vitamin D. A deficiency of vitamin D also interferes with the T-cell mediated immunity.

- Vitamin E is the generic term for tocopherols and tocotrienols
- Food Sources of Vitamin E
- Vitamin E is present in almost all foodstuffs. It is found in wheat germ, corn, nuts, seeds, olives, spinach, asparagus and other green leafy vegetables and vegetable oils like groundnut, soy.

- Absorption and Storage of Vitamin E
- Absorption of vitamin E from the intestine depends on adequate pancreatic function, biliary secretion and micelle formation (A micelle is a spherical cluster of amphipathic molecules (like soap or lipids) in a liquid, forming a tiny particle with a water-loving (hydrophilic) outer shell and a water-hating (hydrophobic) core).
- Emulsification takes place initially in the stomach and then in the small intestine in the presence of pancreatic and biliary secretions.
- Solubilization within mixed bile salt micelles, uptake by enterocytes and secretion into the circulation.

- Functions of Vitamin E
- Vitamin E is the major lipid-soluble antioxidant in the cell antioxidant defense system and is exclusively obtained from the diet.
- The main role of vitamin E and the biological activity of tocopherols are due to its antioxidant property. This antioxidant property of vitamin E is useful for various body processes and substances.

- Protection of erythrocytes: Vitamin E protects erythrocytes from haemolysis by the production of oxidizing agents
- Protection of cell membrane: It protects the cell membrane from getting damaged from naturally occurring peroxides and toxic free radicals formed from fatty acids and oxidative tissue damage as described above.
- Protection against poisoning: It protects liver from injury due to carbon tetrachloride poisoning.

- Protection of both vitamin A and carotene: It protects vitamin A and p-carotene from destruction by oxidation.
- Synthesis of enzymes and proteins: It serves as a co-repressor in the synthesis of certain enzymes and plays a specific role in the synthesis of haem proteins.

- Prevention of diseases: Because of its anti-oxidant function and its role in inhibiting cell proliferation of smooth muscles, vitamin E can be used for prevention and treatment of diseases.
- It has also been suggested that vitamin E supplementation (200- 400 mg/ day) may be appropriate therapeutically to moderate some aspects of degenerative diseases such as Parkinson disease, reduce the severity of neurologic disorders.

- Recommended Dietary Allowances for Vitamin E
- The requirements for the vitamin E are expressed in terms of tocopherol equivalents (TE) as mentioned earlier - 8 mg for females and 10 mg for males

- Health aspects

Deficiency:

- Vitamin E deficiency is a cause of the impaired neuromuscular function, sometimes seen in patients with disorders that interfere with absorption or transport of the vitamin.
- Symptoms include poor reflexes, impaired locomotion, decreased sensation in the hands and feet, and changes in the retina.

Toxicity:

- Adverse effects such as muscle weakness, fatigue, nausea, diarrhea, double vision, elevation of serum lipids, impaired blood coagulation and reduction of serum thyroid hormones occur due to indiscriminate ingestion of excessive amounts of vitamin E over long periods of time.